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Blind Medical Image Watermarking Based on LBP–DWT for Telemedicine Applications

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Abstract

With the increasing use of telemedicine in healthcare, the security and integrity of medical images during transmission have become critical. This paper presents a novel blind watermarking scheme Local Binary Pattern–Discrete Wavelet Transform (LBP–DWT) for medical images based on Local Binary Patterns and the Discrete. Wavelet Transform in frequency domain. We take advantage of the LBP, which is computationally fast, to improve the watermark's resistance to the different kinds of attacks, and maintain the overall visual quality of the watermarked images. In addition, the DWT could offer a high trade-off between robustness and imperceptibility due to the multi-resolution analysis it provides. During embedding, the *LL* band (approximation coefficients) of the DWT is selected and divided into 3×3 blocks. The resulting LBP codes are then XORed with the embedding bits and hidden in the corresponding blocks using the Least Significant Bit technique. Note that the Arnold transform is used during the embedding step to scramble the watermark, which is then vectorized based on the ZigZag fashion to improve the security of the proposed scheme. To evaluate the performance of the proposed method, extensive experiments are conducted on a dataset

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of medical images. The watermarked images are tested against various attacks, including compression, noise addition, and cropping. The obtained results demonstrate the effectiveness of the proposed techniques.

Keywords Watermarking · Blind · Telemedicine · Discrete wavelet transform · Local binary patterns

1 Introduction

Telemedicine refers to the task of providing medical diagnosis, treatment, and consultation services over a secure and integrated telecommunications network. In a digitally connected healthcare ecosystem, healthcare professionals can connect remotely with patients, exchange health information, and deliver care services. Therefore, such a practice will bridge the geographical and temporal gaps and improve healthcare access, cost-effectiveness, and patient diagnosis outcomes. Moreover, telemedicine minimizes the danger of disease transmission by reducing face-to-face interactions. It guarantees that patients receive essential medical attention while assisting in the maintenance of social distance rules. Recent years have seen telemedicine become increasingly important and crucial, especially in light of the COVID-19 pandemic.

Digital watermarking has been emerged to secure the medical exchanges through networks in the telemedicine scenario. The process of watermarking involves carefully inserting information (also known as watermark) into an image. Note that this watermark, which commonly alters the image irreversibly, should be unseen and secret, and its insertion keeps the image's visual quality maintained, especially when it comes to medical imaging, where small details could significantly affect the diagnosis results. Furthermore, the watermarking system should be robust enough against the different kinds of attacks such as the Gaussian filtering, Gamma correction, and geometrical attacks. Thus, a good trade-off between imperceptibility and robustness is highly recommended. By embedding a hidden watermark inside the image, the watermarking technique overcomes steganography's limitations [24].

The current studies on image watermarking can roughly be categorized into spatial and frequency-based approaches. In the spatial domain approaches [30], the image pixels are directly modified to insert the watermark. For instance, the Least Significant Bit method (LSB) can be considered as the most straightforward spatial-based method with a high embedding capacity [24], where each pixel's rightmost bits are replaced with the values of the watermark. As for the frequency-based methods, the watermark is hidden in the image transform. A multitude of transforms are considered for this end including DiscreteeCosine Transform (DCT) [4]. DWT [38] and the Singular Value Decomposition (SVD) [7]. While the spatial-domain techniques are computationally faster than the frequency domain, they are often less robust against geometric attacks. In addition, concealing the watermark within the transformed coefficients will produce an image with a small change in the pixel correlations, thereby maintaining a higher level of visual quality [28]. According to the watermarking strategy, the existing techniques can be classified to blind [10], semi-blind [32] and non-blind [29]. While the blind methods do not require the original image to extract the watermark, the non-blind

methods need to access the original image to recover the watermark. In the semi-blind approach, extra information is required to retrieve the watermark [25].

In this paper, we propose a blind watermarking scheme for securing images exchanged in the telemedicine process based on combining the Local Binary Patterns (LBP) image features and the frequency sub-bands of the DWT. The LBP is a computationally fast method used for texture analysis [23] and was originally used for biometrics identification tasks [1, 27]. By harnessing the LBP in this context, we aim to enhance the watermark's resilience to various attacks and transformations, all while preserving the overall visual quality of the watermarked images. As for the DWT, it presents several advantages: (1) it allows analyzing the frequency components in a localized manner, making possible to pick up the most eligible sub-band to integrate the watermark, (2) it provides a multi-resolution analysis of the image by decomposing the image into distinct levels of details, which offer other possibilities to enhance the compromise between the imperceptibility and robustness, (3) it could offer more robustness by inserting the watermark in different frequency sub-bands and scales.

Particularly, the proposed scheme aims at protecting the medical image by integrating the patient information into the sub-bands of the DWT. At first, the DWT is used to decompose the image into four sub-bands namely *LL, HL* (horizontal detail coefficients)*, HH* (diagonal detail coefficients) and *LH* (Vertical detail coefficients). Then, each block's LBP code is created by thresholding the pixel's immediate neighborhood based on the value of the central pixel. The values that are higher than the central pixel value are assigned with ones and the remaining pixels with zeros. The obtained binary code is then XORed with the binary watermark to generate a new encrypted code. To improve the method security, the new code is then shuffled and embedded in the *LL* sub-band using the LSB mechanism. This will allow a slight modification of the bits, thus preserving the medical image quality, and avoiding false diagnoses that could be caused by the image degradation. Note that *LL* sub-band is empirically selected as it provides the best trade-off between robustness and imperceptibility. During the watermark extraction, we utilize a simple yet effective trick to adjust the pixel values in order to maintain the initial LBP codes and assure a faithful retrieval of the integrated watermark. To assess the imperceptibility and robustness of the proposed method, we consider the most common attacks such as compression, noise adding and filtering.

The rest of this paper is organized as follows. In Sect. 2, we review the related works for medical image watermarking. The background of the concepts used in this study is given in Sect. 3. Section 4 presents the details of the proposed method. We dedicate Sect. 5 to present the experimental results. Finally, Sect. 6 draws some conclusions and future works.

2 Related Works

Several attempts have been made to achieve robust watermarking systems for telemedicine tasks. As has already been mentioned, existing studies can roughly be classified into spatial and frequency-based approaches. The spatial approaches have several limitations, among which is the difficulty of achieving a high imperceptibility, as pixel values are directly modified to include the watermark. Another issue with such

type of approaches lies in the weakness against geometric attacks. In the following, we will review some literature works with a focus on the frequency-based approaches.

Alomoush et al. [3] present an inconspicuous watermarking approach designed for embedding data within the transformed domain of grayscale images. This method entails the integration of a concealed message into the LSB of the DCT coefficient, employing a linear modulation algorithm for this purpose. DWT and SVD are combined by Zermi et al. [42] The retinal image is subjected to a DWT first, and then the LL sub-band is subjected to an SVD. After that, the watermark is appended to LSB of the S component, which is produced by adding the parity of consecutive coefficients. The DWT is utilized in [31], and the Whale Optimization Algorithm (WOA) cost function evaluation is used to find the ideal embedding coefficients. SVD is used to embed the watermark in the medical image. A blind image watermarking technique reliant on DWT and SVD is introduced in [33]. In this method, DWT is employed to process the region of interest (ROI) within a medical image, yielding distinct frequency sub-bands. In the LL band of the ROI, block-SVD is used to obtain diverse singular matrices. In the left singular value matrix of these blocks, this procedure finds pairs of elements that have similar values. These similar values are then adjusted using a specified threshold to embed the watermark. Devi et al. [10] employ a GWO-optimized hybrid DWT-DCT-SVD scheme to sequentially embed three binary watermark images into the cover image, one at a time.

Wang et al. [39] introduce a fragile watermarking technique for image authentication that utilizes the local binary pattern (LBP) and DWT. Within this approach, the LBP pattern is applied to the low-frequency wavelet coefficients, serving as a distinctive watermark feature. However, it should be mentioned that the authors have considered the fragile scenario, as these features are then inserted into the LSB of the maximum pixel value within each block of the host image. Furthermore, the positions of the maximum pixel values are pre-stored for subsequent use in watermark extraction. Verma et al. [36] use a hybrid methodology for medical image watermarking that combines DWT and SVD. They start by extracting different frequency sub-bands from the image using DWT. They then process the LL sub-band using the SVD. A scaling factor is used to insert the watermark into the S-value in order to integrate it. A modified LL sub-band is produced when the original U and V parts are combined with the modified S part. In order to obtain the watermarked image, they applied an inverse DWT on modified LL sub-band. Anand et al. [6] propose a method for embedding multiple watermarks in medical images, consisting of text and image. They start by decomposing the host image into four sub-bands using DWT. The text is then embedded into the singular value S using SVD on the sub-band HH. After applying SVD to both sub-bands, the watermark is divided into two parts and combined with S of the HL and LH sub-bands of the cover image. Kahlessane et al. [18] present a blind watermarking technique for seamlessly integrating electronic patient records into computerized tomography scans. The approach begins with the application of a DWT to the image. Then, the ZigZag scanning technique is applied to perform a topological reorganization of the LL sub-band coefficients. Next, the obtained coefficients are merged to incorporate the watermark segments, encompassing a hashed integrated electronic patient record. Falih et al. [26] present an enhanced fragile watermarking approach that employs a modified LBP, specifically utilizing local pixel contrast for

watermark embedding. In this approach, this method uses an enhanced LBP operator on the host image to distinguish between smooth and rough blocks. Subsequently, the LSB is utilized to insert watermark bits into the host image. During this process, 2–3 bits are inserted in the rough blocks, while 1–2 bits are inserted in the smooth blocks, enabling watermark extraction. In Devi et al. [11], the process of dividing the image into blocks allows the watermark to be integrated into the host DICOM image. A DWT is applied to every block. For every block in this transformation, a particular set of DWT coefficients is selected randomly using a cryptographic key. Then, using a rank-based criterion as guidance, the encrypted watermark is embedded into these chosen coefficients.

Deep learning-based watermarking has gained significant attention due to its applications and impact on image security [12]. It offers several advantages, such as identifying optimal embedding positions and determining the best embedding strength while preserving robustness and imperceptibility. Deep learning also helps reduce noise in the watermark and can simulate attacks to extract it more effectively [16]. As a branch of artificial intelligence, deep learning uses neural networks to mimic the human brain's processing and decision-making, extracting image features and adaptively embedding watermarks for superior results. Although research in this area is still in its early stages, deep learning-based watermarking models, such as GANs [13], DNNs [14], and CNNs [22], are becoming more common. These models typically include convolutions, fully connected layers, pooling layers, and gates, and platforms like Keras, Torch, and TensorFlow make it easier to build and train robust neural networks for various applications.

Despite the advancements in deep learning for watermarking, traditional methods still remain feasible and valuable for several reasons. First, traditional methods typically require less computational power and do not need extensive training datasets, making them more accessible and easier to implement, especially in resource-constrained environments [41]. Furthermore, these methods, such as those based on DWT or LBP, are often more interpretable, which makes it easier for researchers and practitioners to understand, modify, and debug the algorithms. Additionally, traditional techniques can be highly efficient for specific applications, such as medical image watermarking, where the focus is on embedding small, robust watermarks without complex computations[37]. Moreover, traditional methods are less susceptible to issues like overfitting, which can occur with deep learning models when trained on limited data or exposed to out-of-distribution images.

A review of existing image watermarking techniques reveals a persistent need for improvement, particularly in balancing imperceptibility, capacity, and robustness. Current methods often prioritize one aspect at the expense of the others. This paper proposes a novel approach that leverages well-established machine learning techniques like Local Binary Patterns (LBP) to optimize embedding strategies and achieve a better balance between these crucial watermarking properties.

3 The Work Background

3.1 Local Binary Pattern (LBP)

LBP is a widely used descriptor in a variety of domains, such as texture analysis [17, 18], image authentication, and image forgery identification. Its many uses are facilitated by its efficacy as a texture feature describer [19, 20] and in medical diagnosis [2].

In the LBP procedure, the image is partitioned into multiple sub-blocks, with a specified size of n × n, illustrated by the example of m = 3 in Fig. 1a. The central pixel value serves as a threshold for determining neighboring pixel values. Pixels with values smaller than the threshold are assigned 0, while those equal to or greater than the threshold are assigned 1. The binary values are then considered in a clockwise order and transformed into decimal form. The formulation of LBP is depicted in Eq. 1. Figure 1 illustrates the sample block operator of a 3 × 3 block, along with the conversion of LBP from binary to decimal.

$$LBP_j = \sum_{i=0}^{i=7} S(P_c,\ P_i) * 2^i \tag{1}$$

where P_c represents the value of the central pixel in block j, and Pi represents the corresponding pixel values. The sign function, denoted as S and defined as (Eq. 2), is expressed as:

$$S(P_c,\ P_i) = \begin{cases} 1, & if \quad P_c \geq P_i \\ 0, & else \end{cases} \quad \text{Where:} \quad i \in \{1, \ldots, 8\} \tag{2}$$

Due to its manner in representing image texture, any manipulation of the image will inevitably alter its Local Binary Pattern (LBP). Consequently, the LBP operator serves as a effective tool for detecting image tampering [34]. In the proposed algorithm, the LBP operator is employed to produce authentication embedding bits directly from the host image, which will be XORed with the watermark. This approach facilitates blind tamper detection, allowing for the identification of unauthorized alterations without prior knowledge of the original content.

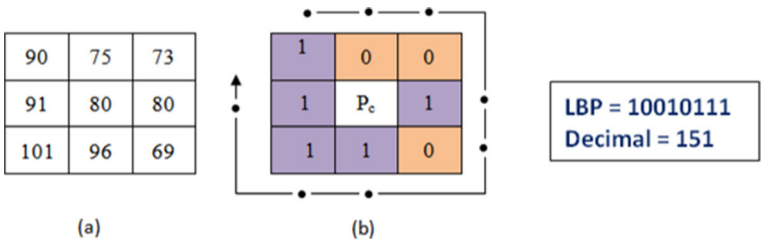

Fig. 1 LBP Operator on a 3 × 3 block: **a** original image block, **b** corresponding LBP

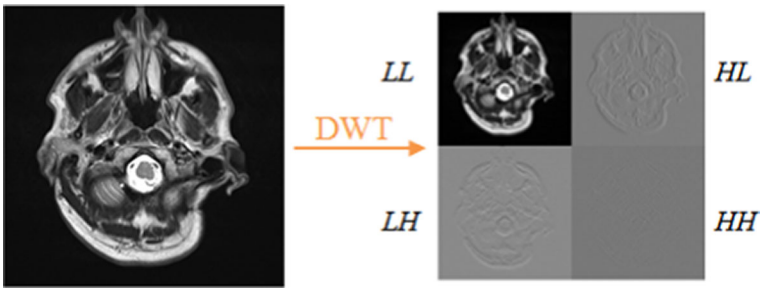

Fig. 2 Decomposition shown at the 1-level discrete wavelet transform

3.2 The Discrete Wavelet Transforms (DWT)

The DWT is a technique that makes it straightforward to create orthonormal wavelet bases. Its fundamental concept is to decompose the signal being studied into a set of approximation details and coefficients. Subsequently, every approximation series is splitted into additional sets of approximation and detail coefficients. The wavelet properties of the signal at a given scale are represented by the detail coefficients at each stage of this segmentation. An image is divided into 4 sub-bands by the DWT: three sub-bands for the vertical (LH), horizontal (HL), and diagonal (HH) detail coefficients, as well as an LL sub-band with approximation coefficients [10] (Fig. 2). The coefficients within each sub-band are computed using the Haar filter as follows:

$$LL(x, y) = \frac{p(x, y) + p(x, y+1) + p(x+1, y) + p(x+1, y+1)}{2} \tag{3}$$

$$LH(x, y) = \frac{p(x, y) + p(x, y+1) - p(x+1, y) - p(x+1, y+1)}{2} \tag{4}$$

$$LH(x, y) = \frac{p(x, y) - p(x, y+1) + p(x+1, y) - p(x+1, y+1)}{2} \tag{5}$$

$$HH(x, y) = \frac{p(x, y) - p(x, y+1) - p(x+1, y) + p(x+1, y+1)}{2} \tag{6}$$

3.3 Arnold Transform

In our proposed watermarking scheme, we incorporate the Arnold Transform in order to improve our watermarking system's security. The general Arnold Transform can be described as follows:

$$\left(\frac{i\prime}{j\prime}\right) = \begin{pmatrix} 1 & a \\ b & ab+1 \end{pmatrix} \left(\frac{i}{j}\right) (mod\, L) \tag{7}$$

In this equation (Eq. 7), (i, j) represents the original pixel positions in the image, (i', j') are the positions that correspond to the following permutation. The control parameters a and b allow for pixel position adjustments, and L represents the image size.

For different image sizes and parameters, The period T in the Arnold transform varies for different image sizes and parameter settings, where T depends on the image size [21],and it determines the number of permutations required to return the image pixels to their original positions.

The use of the Arnold transform not only scrambles the image but also provides enhanced security through the use of the T value as a key in our scheme [25].

4 The Proposed Method

The majority of watermarking algorithms share a common underlying model, with variations primarily occurring in specific strategies during the watermark insertion or detection processes [37]. In this paper, we introduce an improved blind and robust watermarking method by using DWT and LBP which is used as a map on the top of the obtained sub-band. The proposed method has two stages, watermark embedding and extraction phases which are described hereafter.

4.1 Watermark Embedding

In this step, the medical image is normalized and transformed to the frequency domain by *DWT* to obtain sub-bands (*LL, LH, HL, HH*). To investigate the impact of Local Binary Pattern (LBP) on distinct sub-domains, we separately applied LBP to each sub-domain. The outcomes, as presented in Table 1, indicate that the Peak Signal to Noise Ratio (*PSNR*) values are 44.23, 50.07, 50.23, and 51.33 for the *LL, LH, HL,* and *HH* sub-domains, respectively. However, when subjected to various attacks (such as jpeg compression, among others, as listed in the Table 1), it becomes evident that the *LL* sub-band is the most dependable (the values of *NC* and *BCR* are the highest).

Table 1 Reason for selecting of sub-band

	Metrics	LL	LH	HL	HH
Watermarked MI	**PSNR**	44.23	50.07	50.23	51.33
	SSIM	0.9894	0.9921	0.9928	0.9969
Jpeg compression (q = 50)	**NC**	0.9780	0.6797	0.6471	0.9633
	BCR	0.9685	0.6824	0.6736	0.5540
Jpeg compression (q = 60)	**NC**	0.9936	0.6480	0.6929	0.8176
	BCR	0.9905	0.7075	0.7239	0.5947

Bold indicates the highest NC and BCR values after the attack

After determining the suitable sub-band (LL), we will clarify the procedures for implementing LBP in this specific sub-band:

Step 1: The *LL* sub-band is partitioned into 3×3 non-overlapping blocks, and subsequently, each block is multiplied by 10. The reference coefficient for each block is represented by P.

Step 2: considering the image patch presented in Fig. 3 as one of the *LL* blocks (P), the embedding of eight bits W_i is performed using the ensuing steps.

Step 3: We split block P into two blocks: one, PI, contains integer values, while the other, PF, holds fractional values (Fig. 4).

Step 3: The 8 bits of LBP code are extracted from PI using Eqs. (1) and (2).

Step 4: The watermark W is transformed into the watermark W^A by applying the Arnold transform with a secret key, to achieve scrambling.

Step 5: We convert the W^A into 1D vector using the ZigZag scan method. Then, we divide the watermark vector into 8-bits segments (Fig. 5).

Step 6: The eight bits W_i^A (Segment i) are encrypted with LBP code using the XOR operation to get (X_j) code using Eq. 8.

$$X_j = W_i^A \oplus LBP_j \tag{8}$$

The *XOR* operation combines the LBP information with the corresponding bits of the watermark to embed the result within the *LL* sub-band.

Step 7: For increased security, the 8 bits of X_j code are arranged into four pairs, each consisting of two neighboring bits. Each pair consists of two bits that are switched

Fig. 3 Representation of a block P (3×3) of obtained sub-band *LL*

P_1 =7,455	P_2 =6,982	P_3 =6,858
P_8 =6,756	P_c =6,845	P_4 =6,126
P_7 =5,458	P_6 =5,492	P_5 =6,328

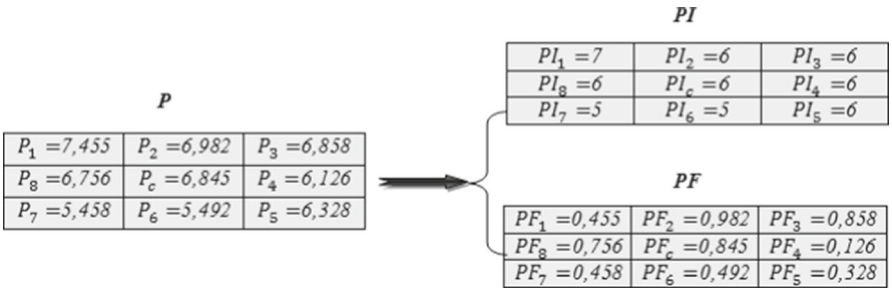

Fig. 4 Illustration of a block P divided into two blocks

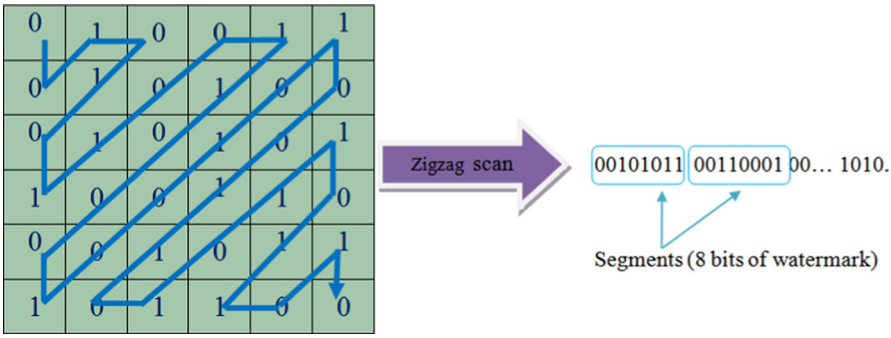

Fig. 5 Illustration of ZigZag scan and segmentation of Watermark

around. Equation 9 yields the resulting Y_j code, as follows:

$$Y_j = \left\{ y_j^0 = x_j^1; y_j^1 = x_j^0; y_j^2 = x_j^3; y_j^3 = x_j^2; y_j^4 = x_j^5; y_j^5 = x_j^4; y_j^6 = x_j^7; y_j^7 = x_j^6 \right\} \tag{9}$$

Step 8: The bits that are present in the Y_j are embedded as follows:

Apply the LSB from the corresponding bits of the code Y to each coefficient of PI block, to obtain watermarked block PI'.
In order to preserve the local neighborhood relationship within the watermarked sub-band, the coefficient values obtained PI' are modified using Eq. 10. This modification ensures that the resulting LBP code remains consistent with the original code (Fig. 6).

$$\begin{aligned} PI_i^* &= PreserveLBP\left(PI_c,\ PI_i,\ PI_i'\right) \\ &= \begin{cases} PI_i' + 2,\ if(PI_c \geq PI_i')AND(PI_c < PI_i') \\ PI_i' - 2,\ if\left(PI_c < PI_i'\right)AND(PI_c \geq PI_i') \\ PI_i' otherwise \end{cases} \end{aligned} \tag{10}$$

Step 9: Merge the two blocks ($PI\prime$ and PF), to obtain the watermarked block P^* (Fig. 7).

Step 10: After changing the sub-band with above mentioned algorithms, we divide each pixel of it by 10 to get the modified *LL*. An inverse DWT is then applied to the modified LL subband and the original detail parameter sets (*HL*, *LH*, and *HH* sub-bands) to get the Normalized Watermarked Medical Image *NWMI*.

Step 11: Ultimately, generate the watermarked image by performing the denormalization process to the NWMI in order to obtain Watermarked Medical Image WMI.

Figure 8 provides a visual representation of the watermark embedding process.

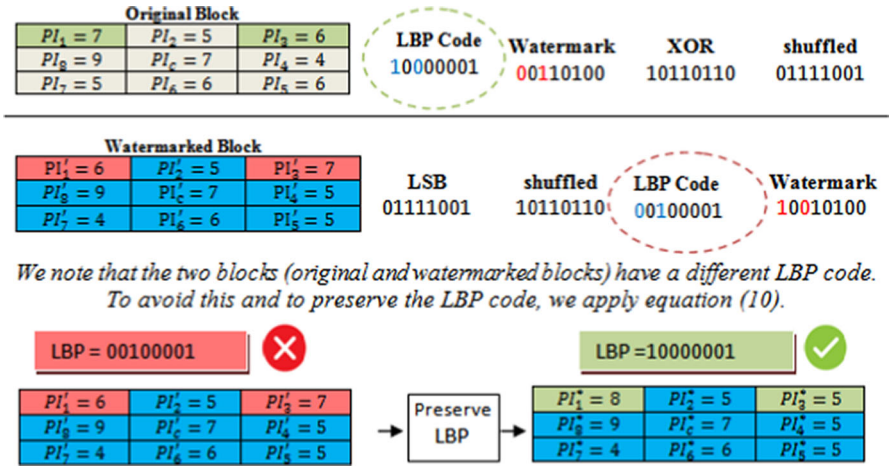

Fig. 6 the preservation process of Block (Eq. 10)

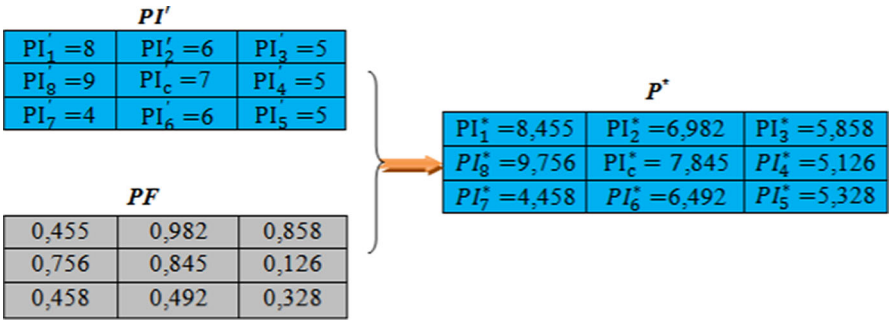

Fig. 7 Merge the two blocks

4.2 Watermark Extraction

In contrast to the watermark embedding process, the watermark extraction process works in reverse. The steps involved in the extraction process are as follows:

Step 1: To obtain the sub-band LL, the image is first normalized, a DWT is applied, and then the image is multiplied by 10.

Step 2: The watermarked sub-band is divided into blocks P_i of size 3×3, as depicted in (Fig. 3).

Step 3: Split block P into two blocks: one, PI, contains integer values, while the other, PF, holds fractional values (Fig. 4).

Step 4: Similarly, for each block, The calculation of the Y_j as follows:

1. For each block PI, the 8-bits of LBP code are extracted using Eqs. (1) and (2).
2. Extract the LSB from the coefficient of modified sub-band by analyzing the eight coefficients surrounding the central coefficient to obtain Y shuffled code.

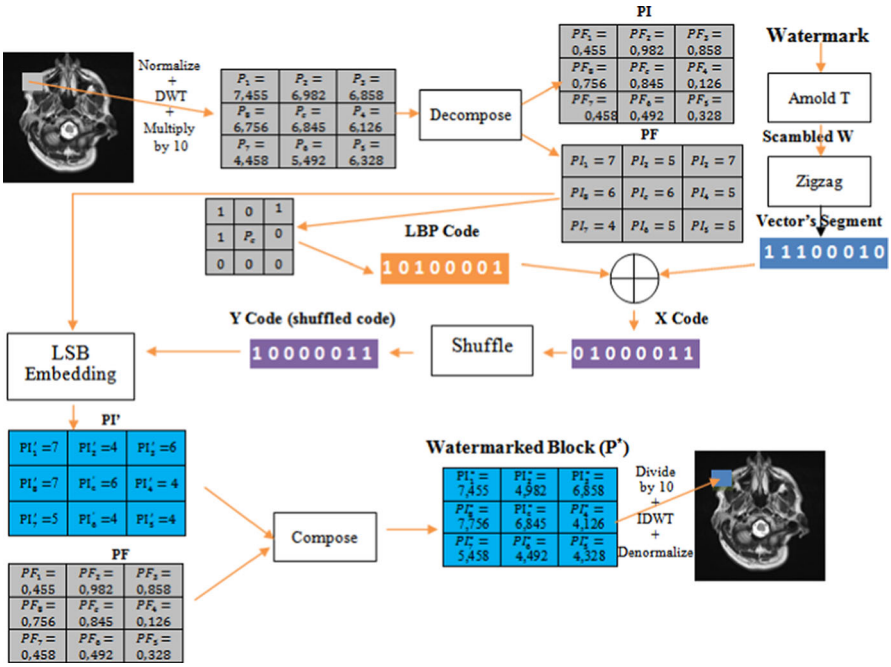

Fig. 8 Embedding algorithm of the proposed method

Step 5: The pixel X_j are recovered from shuffled code Y_j using (Eq. 11).

$$X_j = \left\{x_j^0 = y_j^1; x_j^1 = y_j^0; x_j^2 = y_j^3; x_j^3 = y_j^2; x_j^4 = y_j^5; x_j^5 = y_j^4; x_j^6 = y_j^7; x_j^7 = y_j^8\right\} \tag{11}$$

Step 6: Next, an XOR operation between the X code and the LBP code is used to extract 8 physical bits from the watermark sequence (Eq. 12).

$$W_i^A = X_j \oplus LBP_j \tag{12}$$

Step 7: The extracted bits from the watermark sequence (segment) are concatenated to form the complete watermark vector.

Step 8: Apply the inverse ZigZag scan operation to the modified vector, as done in the fifth embedding step, to obtain the matrix of a modified LL.

Step 9: Finally, inverse Arnold transform with a secret key is applied to W^A to get extracted watermark EW.

A graphic depiction of the watermark extraction procedure is given in Fig. 9.

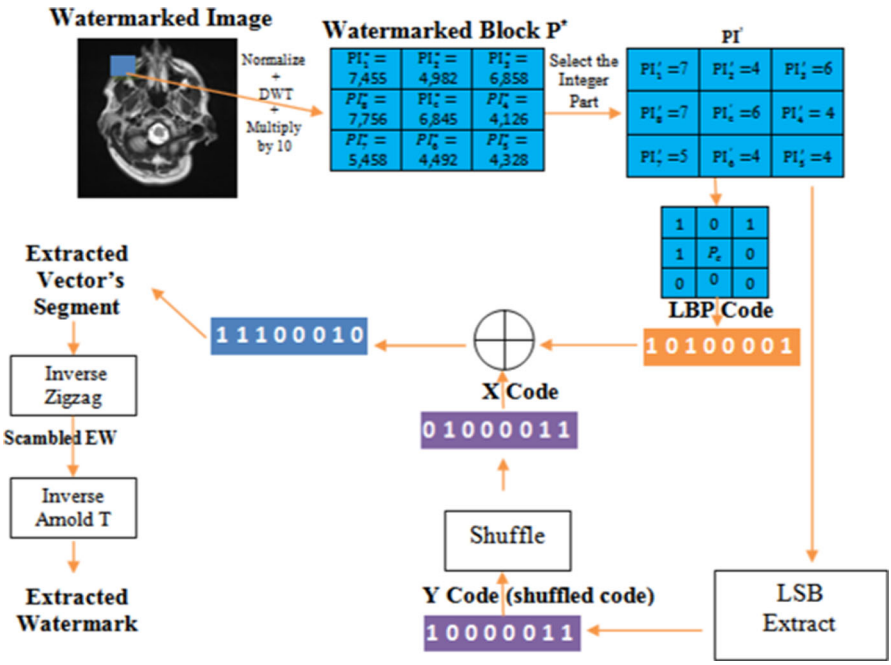

Fig. 9 The process of extracting the watermark's eight bits

5 Experimental Results and Discussion

5.1 Experimental Setup

Numerous medical image types, such as X-ray, CT, ultrasound (US), and MRI, have been used to evaluate the suggested watermarking scheme. To facilitate analysis, alphanumeric labels are used to represent the image names. These medical images are in a format of 512 × 512 pixels as illustrated in Fig. 10. As for the watermark, we **have** converted the patient's information to a binary image and use it as a text-based watermark. This is commonly used for copyright and ownership purpose (Fig. 11).

Google Colab software is employed to implement the proposed method. A total of twelve (12) sample medical images obtained from various Kaggle datasets [Ultrasound Images Dataset, CT and MRI brain dataset, Xray dataset] were used in our experiments. To comprehensively assess the effectiveness of the proposed watermarking method, several attacks are applied to it. PSNR and SSIM are the metrics employed for evaluating perceptual attributes. PSNR is specifically used to quantify the visual similarity between the original image and the watermarked image. Following watermark embedding, both the original and watermarked images should exhibit remarkable visual similarity, with only minor distinctions between them.

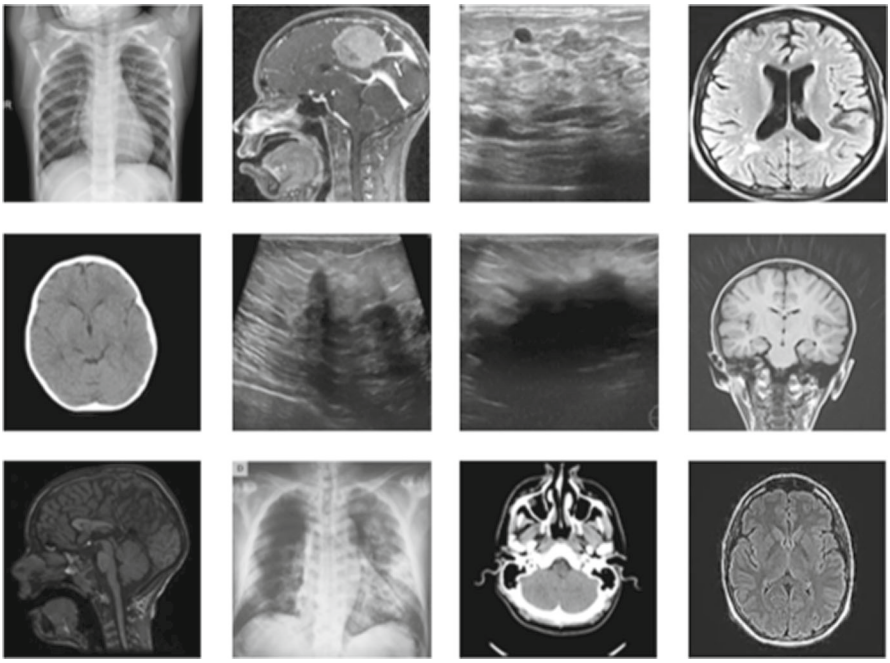

Fig. 10 X-ray, CT, US, and MRI host medical images utilized in our suggested method

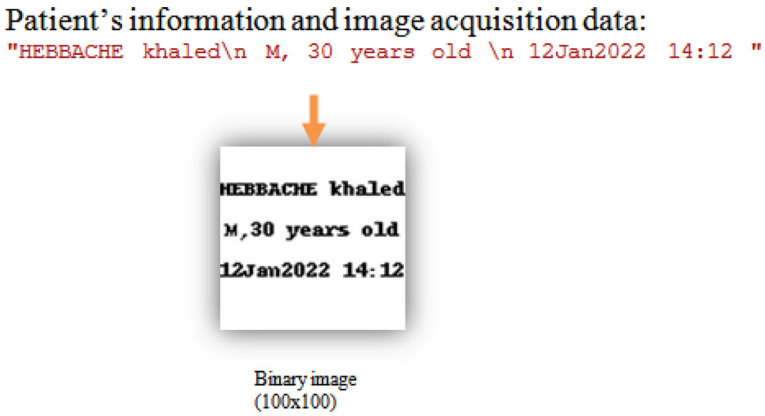

Fig. 11 Creation of watermark

The PSNR, as described by [21, 40] quantifies how much the original image has degraded in dB due to the integration process:

$$\mathrm{PSNR_{dB}} = 10\ \log_{10}\left[\mathrm{N} * \mathrm{M}\frac{\max \mathrm{I}^2(i,\ j)}{\sum i,\ j[\mathrm{I(i,j)} - \mathrm{J(i,\ j)}]^2}\right] \tag{13}$$

Here, I and J represent the original image and the watermarked image, respectively, both having sizes N × M.

A metric for assessing the similarity between two images is SSIM [9]. It determines the separation between two N x N segments by calculating over several image windows:

$$\mathrm{SSIM}(x, y) = \frac{\left(2\mu_x\mu_y + C_1\right)\left(2\delta_{xy} + C_2\right)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\delta_x^2 + \delta_y^2 + C_2\right)} \tag{14}$$

The symbols represent the following: μ_x: The average of x. μ_y: The average of y. δ_x^2: The variance of x. δ_y^2: Variance of y. δxy: Covariance between x and y.

BCR (the bit-correct ratio) and NC (Normalized Correlation) values are estimated to verify the quality of extracted watermark EW with the original watermark W using Eqs. 15 and 16:

A correlation coefficient, denoted as NC [15], indicates how similar the original image and the one under attack are. Two images are highly identical if the coefficient (NC) is 1. If the coefficient (NC) is 0, two images are completely different. This coefficient ranges from 0 to 1. Equation 15 is utilized to calculate this metric.

$$NC(w, w_e) = \frac{\sum_{p=1}^{M}\sum_{q=1}^{N}(w(i, j) - \overline{w}(i, j)(w(i, j) - \overline{w_e}))}{\sqrt{\left(\sum_{p=1}^{M}\sum_{q=1}^{N} w(i, j) - \overline{w}^2\right)\left(\sum_{p=1}^{M}\sum_{q=1}^{N} w_e(i, j) - \overline{w_e}^2\right)}} \tag{15}$$

where w(p, q) and w_e(p, q) denote the intensities of the pixel at coordinates (p, q) in the original w and the extracted watermark w_e, respectively. The averages of intensities for the watermark image *w* and the extracted watermark w_e are denoted by $\overline{w}$ and $\overline{w_e}$, respectively.

The Bit Error Rate (BER) assesses the number of errors between the extracted watermark and its original, while the Bit Correct Rate (BCR) assesses the number of corrects between the extracted watermark and its original [5, 22]. The following expression for its estimation is given by Eq. 16:

$$BCR = 1 - BER = \frac{\text{Number of correctly decoded bits}}{\text{Total number of embedded bits}} \tag{16}$$

5.2 Experimental Results

To ensure robustness, we consider using the DWT in the frequency domain along with the LBP in the spatial domain to maintain the imperceptibility. This approach effectively blends the spatial and frequency domains to balance the trade-off between robustness and imperceptibility.

Figure 12 presents the outcomes of using the system against CT, MRI, X-ray, and US scans.

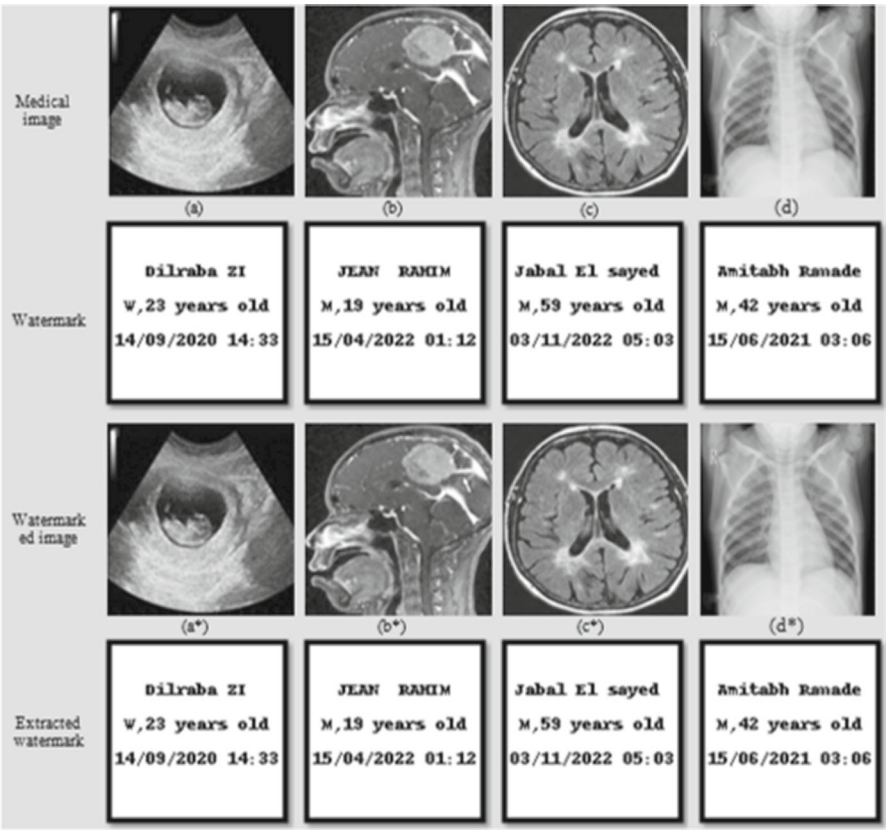

Fig. 12 Exemple illustrate watermarked MI and its extracted watermark

5.3 Imperceptibility Evaluation

Medical image watermarking must carefully maintain image quality while integrating sensitive patient data. This is because watermark insertion process may introduce distortions to the medical image. Thus, to assess the imperceptibility of our suggested approach, we measure the image quality after inserting the watermark using the metrics defined in Sect. 5.1 i.e., PSNR and SSIM.

Figure 13 presents the impercibility results in terms of PSNR and SSIM.

From the results we obtained, it is clear that the PSNR and SSIM values significantly exceed the threshold of 44 dB and 98%, respectively. This means that embedding the LBP code using LSB does not affect the image and keeps the imperceptibility value high. These high values indicate that the original *MIs* and their corresponding *WMI*s show close visual similarity.This indicates that the visual systems of humans are unable to distinguish differences between host *MI* and their corresponding *WMI* due to their extreme similarity. These high PSNR and SSIM values affirm that the watermarking

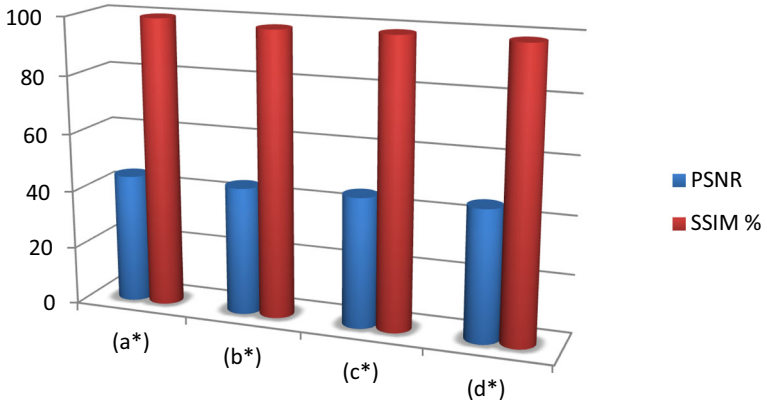

Fig. 13 Imperceptibility results in terms of PSNR and SSIM

techniques preserve the visual quality of *MIs* with an imperceptible impact, which is crucial for maintaining the integrity of diagnostic images in the medical field.

5.4 Robustness Tests

In the absence of attack, we use the suggested extraction algorithm to extract watermarks from watermarked medical images. Figure 12 show these extracted watermark from watermarked images including CT scan, MRI, X-Ray and Ultra-sound, and all results are with NC = 1 and BCR = 1. This denotes the precise and accurate extraction of the watermark. So, we can extract accurately watermark images in no attack case. In order to preserve the patient's information, the watermarking algorithm must resist to the various attacks. In order to verify the robustness of the proposed algorithms, we test our algorithm against various types of attacks, including JPEG compression, Gaussian noise, Salt and Pepper noise, median filter and other attacks. The NC and BCR are then calculated between the original and extracted watermark. In the next subsections, we provide more details on the performance of our method against different kinds of attacks.

5.4.1 Robustness Against Image Compression

In Fig. 14 illustrates the BCR and NC values for the watermarks extracted after subjecting them to a JPEG compression attack. The Quality Factor (Q) in JPEG compression refers to a parameter that controls the trade-off between image quality and file size. It is a numerical value (usually ranging from 1 to 100) that determines the level of compression applied to the image. In this case, the quality factor (Q) was adjusted within the range of 10 to 70 [8]. From Fig. 14, it appears that, for all the types of medical images, the watermark is successfully restored under strong JPEG attacks (with BCR = 1 and NC = 1) when the quality factor (Q) exceeds 70. Figure 15 shows

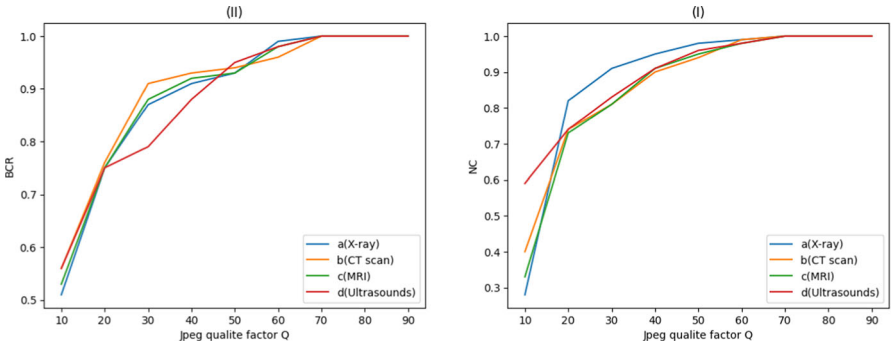

Fig. 14 Robustness against compression attacks. (I) NC curve by Q. (II) BCR curve by Q

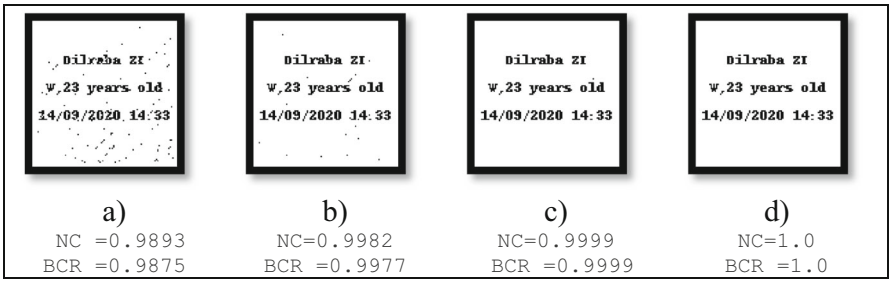

Fig. 15 The extracted watermark according Q: **a** Q = 50, **b** Q = 60, **c** Q = 70, **d** Q = 80

how the extracted watermark transforms as the factor (Q) changes in response to a JPEG compression attack.

Despite the change in the value of Q, we can note that the readability of the extracted watermark initiates at Q = 50, as shown by Fig. 15. Furthermore, Fig. 14 indicates that both NC and BCR tend to approach a value of 1 when Q surpasses 60. These observations from the two figures affirm the robustness of our method when subjecting a watermarked medical image to a JPEG compression attack.

5.4.2 Robustness Against Image Processing Attacks

In the following analysis, we assess the effectiveness of the proposed method when subjected to various image processing attacks, including noise addition, low pass filtering (Average, Gaussian, Median), gamma correction, histogram equalization, Laplacian sharpening, and bit-plane removal. In the bit-plane removal attack, the least significant bits of the watermarked image are substituted with zeros. Table 2 displays the BCR and NC values for the extracted watermarks after these attacks. While smoothing filters like average and Gaussian filters (3 × 3) have minimal impact (BCR and NC values remain high across all image types), a larger Gaussian filter (5 × 5) reduces effectiveness for CT scans (BCR drops from 0.9760 to 0.9367) and ultrasounds. Similar trends hold for the median filter. Salt and pepper noise has a moderate effect – low levels

Table 2 BCR and NC values of the proposed technique under image processing attacks

Attacks	A (CT scan)		B (X-ray)		C (MRI)		D (Ultrasounds)	
	NC	BCR	NC	BCR	NC	BCR	NC	BCR
Average filter (3 × 3)	0.9678	0.9533	0.8177	0.7866	0.9687	0.9545	0.9353	0.9169
Gaussian filter (3 × 3)	0.9833	0.9760	0.9003	0.8792	0.9800	0.9817	0.9646	0.9580
Gaussian filter (5 × 5)	0.9524	0.9367	0.7761	0.7397	0.9575	0.9421	0.9199	0.8987
Median filter (3 × 3)	0.9666	0.9523	0.9108	0.8794	0.9174	0.9265	0.9028	0.9094
Salt and pepper noise (var = 0.01)	0.9691	0.9636	0.9679	0.9646	0.9524	0.9524	0.9703	0.9696
Salt and pepper noise (var = 0.02)	0.9278	0.9252	0.9141	0.9202	0.8916	0.8862	0.8290	0.8276
Histogram equalization	0.7195	0.2143	0.3528	0.3093	0.9433	0.5920	0.8378	0.5474
Laplacian Sharpening	0.7264	0.5854	0.5217	0.5000	0.9366	0.7551	0.8720	0.7251
Bit-plane removal (5 bits)	0.9575	0.9551	0.4195	0.4250	0.9178	0.9238	0.8349	0.8381
Bit-plane removal (6 bits)	0.9737	0.9609	0.6710	0.6392	0.9737	0.9536	0.9362	0.9019
Gamma correction (1.1)	0.9947	0.9998	0.9976	0.9993	0.9819	0.998	0.9891	0.9909

(variance = 0.01) cause minimal change, while higher levels (variance = 0.02) lead to a decrease in BCR and NC (e.g., X-ray NC falls from 0.9679 to 0.9141). Histogram equalization poses a significant challenge, especially for CT scans and X-rays (BCR drops to around 0.2) due to its impact on intensity distribution. Sharpening filters like Laplacian sharpening also negatively affect CT scans and ultrasounds (BCR drops below 0.7), likely due to modifications in image edges. Interestingly, MRIs display surprising resilience in some cases. Removing a moderate number of bit planes (5 bits) has a moderate impact on CT scans (BCR falls to 0.9575) but minimal effect on MRIs (BCR remains around 0.9). Overall, the interplay between attack type, image modality, and watermark robustness is evident.

Figure 16 demonstrates the successful extraction of the watermark from the watermarked medical images subjected to the different attacks namely Salt and pepper

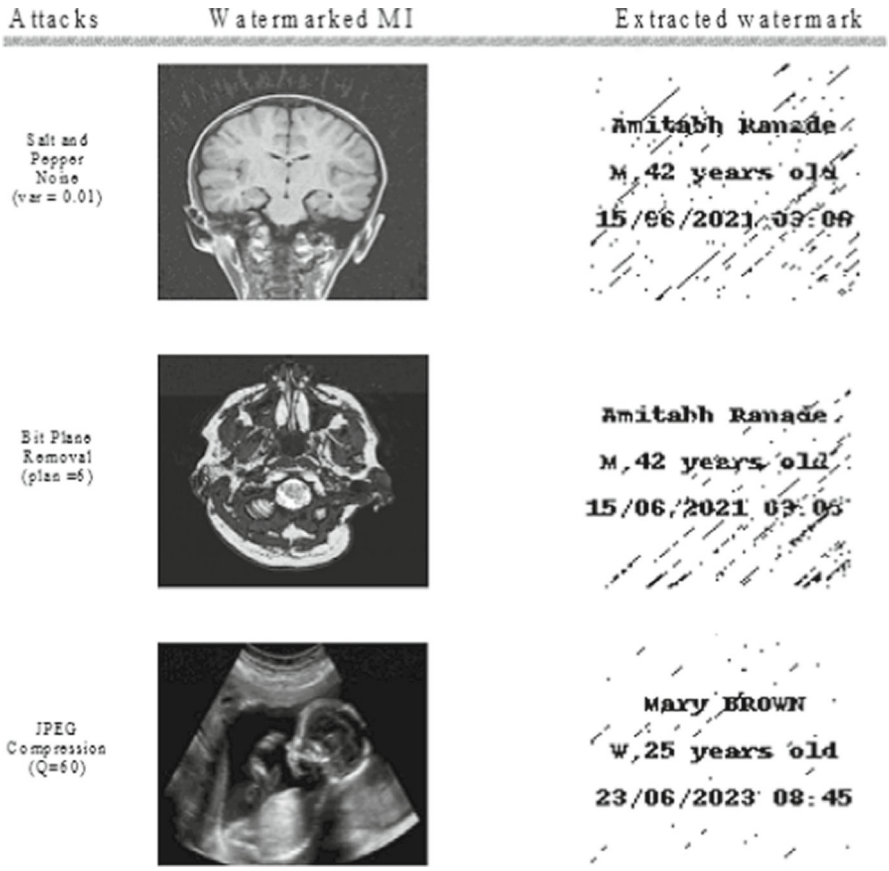

Fig. 16 Attacked images with Gamma (g = 1), Salt and Pepper noise (var = 0.01), LSB removal (6 bits) and JPEG lossy (Q = 60)

noise (var = 0.01), and Bit plane removal (plan = 6). The readability of the extracted watermark highlights the robustness of our algorithms against these attacks.

5.4.3 Robustness Against Geometrical Attacks

The resilience against the geometrical attacks is demonstrated in the following experiments. The suggested approach is comparatively some resistant to geometrical attacks, as shown by Table 3 and Fig. 17. All modalities demonstrate good robustness to resizing (512 → 256 → 512), with NC and BCR values generally above 0.88. Robustness significantly decreases with increasing rotation small angles. Performance declines with increasing rotation angles, while small rotations (0.5°) resulted in a moderate performance drop, larger rotations (1°) led to a substantial decline in both BCR and NC for all modalities. CT scans consistently demonstrated the highest resilience to rotation, while Ultrasound images proved most susceptible. This highlights the vulnerability of the current watermarking scheme to larger rotational distortions.

Table 3 The suggested technique's BCR and NC values under geometrical attacks

Attack	A (CT scan)		B (X-ray)		C (MRI)		D (Ultrasounds)	
	NC	BCR	NC	BCR	NC	BCR	NC	BCR
Resizing (512 → 256 → 512)	0.9787	0.9744	0.9108	0.8884	0.9128	0.9114	0.8949	0.8896
Rotation (0.5°)	0.7804	0.6887	0.6686	0.6037	0.6314	0.6519	0.6343	0.6584
Rotation (1°)	0.7045	0.6143	0.5678	0.5617	0.5734	0.5314	0.5509	0.6097
Surrounding crop (10%)	0.7462	0.6924	0.9754	0.6968	0.9950	0.7195	0.9917	0.7166
Top left quarter crop (0.25)	0.9487	0.7827	1.0000	0.9846	1.0000	0.7864	0.9996	0.7864

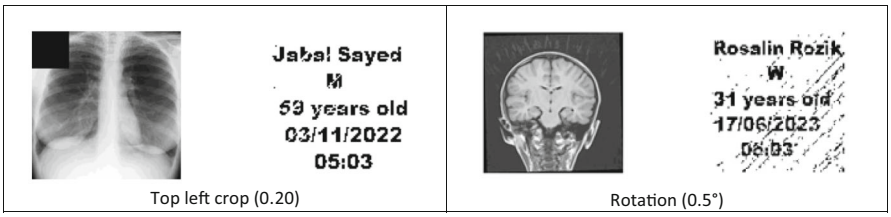

Fig. 17 Attacked Images and Corresponding Extracted Watermarks

Cropping effects also vary—CT scans and ultrasounds generally show lower BCR and NC (around 0.7) with surrounding crops, while X-rays (> 0.9) and MRIs (around 1.0) in some cases (top-left cropping) exhibit minimal impact. These findings emphasize the interplay between attack type, image modality, and watermark robustness.

5.5 Comparison with Other Methods

In this subsection, we compare the performance of our proposed DWT-LBP watermarking method with three other existing approaches for medical image watermarking which are presented in [4, 5, 35].

The approach presented in [5] involves the extraction of the frequency content of the image using a redundant DWT. The final coefficients are then subjected to Schur decomposition, and the watermark bits are inserted by altering the Eigen values' LSB. In [35], the lower resolution approximation module of DWT and the approximation coefficients from LWT are selected for the watermark embedding. Based on LBP features, the scaling and embedding factors are adaptively computed. Concerning the approach presented in [6], text and image watermarks are prepared, the cover image undergoes DWT decomposition, and *HL* and *LH* components are used to embed the medical watermark. The encoded text watermark is placed in the *HH* sub-band of the DWT coefficients.

Table 4 presents a comparison between the proposed scheme and other compared methods in terms of imperceptibility, it can be notice that the our scheme have the highest PSNR and the second highest SSIM (44.23 and 0.9888 respectively),we can therefore conclude from the obtained comparison our approach is guaranteeing a reasonable PSNR and SSIM values. Table 5 reveals that the proposed method has surpassed the compared methods in terms of robustness when facing comparable attacks (the parameters for the attacks are 0.01, 0.01, 60, and 3×3 for Salt and

Table 4 Comparisons of imperceptibility (PSNR & SSIM) among different methods

	[5]	[35]	[6]	The proposed
PSNR	43.28	36.00	36.54	44.23
SSIM	0.9864	0.9884	0.9961	0.9888

Table 5 Comparisons of robustness (BCR & NC) among different methods

Attacks	[5]		[35]	[6]	The proposed	
	NC	BER	NC	NC	NC	BER
Salt and Pepper Noise	0.9755	0.0926	0.9969 (0.001)	0.7751 (d = 0.001)	0.9649 (0.01)	0.0375
Gaussian noise	0.9245	0.2015	0.9941 (0.001)	0.3226 (0.05)	0.9570 (0.01)	0.0513
Speckle noise	–	–	–	0.9687 (0.0001)	0.5142 (0.0001)	0.2858
JPEG compression	0.7662 (q = 20)	0.3162 (q = 20)	0.9997 (q = 90)	0.9904 (q = 60)	0.9970 (q = 60)	0.0081
Median filtering (3 × 3)	0.8735	0.2515	0.9987 (2 × 2)	0.9457	0.9244 (3 × 3)	0.0831
Cropping	0.8913	0.1723	–	0.9337	0.948707	0.0447
Scaling	0.7906	0.2205	0.9999 (20.5)	0.7075(2)	0.8916 (2 0.5)	0.1298
Sharpening	0.8258	0.2234	–	0.6763(0.1)	0.7641	0.3586
Histogram equalization	0.8815	0.2212	–	0.7223	0.7133	0.5803

Pepper noise, Gaussian noise, JPEG compression, and median filter, respectively, instead of 0.001, 0.001, 90, and 2 × 2). However, our method shows some weakness when applying the sharpening and histogram equalization attacks, which is a common weakness shared with other methods. These findings suggest that the amalgamation of the LBP and DWT domains yields better results than employing LBP exclusively within the spatial domain for medical image watermarking.

6 Conclusions

In our study, we investigate a model that integrates LBP, DWT Transform to achieve an efficient blind and robust watermarking approach for medical digital images. Our findings reveal that our method exhibits resilience to multiple attacks, primarily due to the implementation of DWT in the LL sub-band. This sub-band encapsulates diverse features from the original medical image, preserving a substantial amount of information. DWT transform generates a series of sub-bands. As a result, attacks are dispersed throughout all of the sub-bands that are generated from the DWT, and there is very little chance that the attack will alter the value of the pixel it has embedded, and then we apply LBP to the first-level LL sub-band within the wavelet transform and use it as a means to enhance the security of the watermark. While our proposed method outperforms modern approaches in several aspects, it remains susceptible to certain attacks, including sharpening, histogram equalization, and rotation. Addressing these

vulnerabilities constitutes a significant area for future research. As part of future work, we propose to explore DWT with second and third-level wavelet transforms, as well as consider the Shearlet Transform, Curvelet Transform or Contourlet Transform, all of which offer sub-band decomposition for enhanced image analysis within the frequency domain. Additionally, we can investigate the application of LBP to other sub-bands, including HL, LH, and HH, to enhance robustness against specific attack types. Our approach also leaves room for potential enhancements, such as combining DWT with DCT or exploring other advanced transforms.

Data Availability The dataset used during the current study is available in the repository [Ultrasound Images Dataset, Oct. 2023. [Online]. Available: https://www.kaggle.com/datasets/aryashah2k/breast-ultrasound-images-dataset] [CT and MRI brain dataset, Oct. 2023. [Online]. Available: https://www.kaggle.com/datasets/darren2020/ct-to-mri-cgan] [Xray dataset, Oct. 2023. [Online]. Available: https://www.kaggle.com/datasets/fusicfenta/chest-xray-for-covid19-detection].

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